

Khushi Patel

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Mechanical Engineering | Minors: Computer Science & Health Technology

Expected May 2027

- **Coursework:** Dynamics of Mechanical Systems, Engineering Materials, Fluid Dynamics, Electrical & Electronic Circuits, Analog Signal Processing, Data Structures & Algorithms, Database Systems

WORK EXPERIENCE

Abbott

Champaign, IL

Engineering Intern

Jun. 2025 – Present

- Executed manufacturing quality checks on nutrition products, verifying ingredient compliance and process reliability across production stages within the Ingredient Center of Excellence
- Designed and built an ingredient qualification and approval/rejection system adopted by 20+ engineers across the Nutrition Dept., reducing processing time by 50%; validated system against production requirements before deployment
- Developed vendor analytics pipeline in Python to automate supplier risk scoring and spend trend analysis across Abbott's large-scale ingredient portfolio, supporting procurement and supply chain reliability decisions

Speech Perception Laboratory at UIUC

Champaign, IL

Undergraduate Research Assistant

Jan. 2025 – May 2026

- Designed and executed data collection protocols for speech intelligibility experiments across 100+ participants in collaboration with national hearing clinics; performed statistical analysis on acoustic datasets using Python
- Contributed to validation of computational speech perception models by testing ASR pipeline outputs against experimental speech data

LEADERSHIP

American Society of Mechanical Engineers

Champaign, IL

Director of Product Development

May 2025 – May 2026

- Led 50+ member org delivering hardware engineering solutions for 4 industry and academic clients, overseeing design, prototyping, and testing across 4 concurrent medical device, robotics, and embedded systems projects
- Directed a venture-backed hardware startup initiative from concept through prototype; team selected for a 2026 Y Combinator cohort

Intelligent Mattress Topper

Champaign, IL

Lead Engineer

Oct. 2024 – Aug. 2025

- Led design and prototyping of a smart pressure-relief mattress pad integrating pneumatic actuation, a 16-sensor pressure array, and embedded control to improve early pressure ulcer detection by 60%
- Designed and laid out embedded PCBs in KiCAD for microcontroller-driven air-valve actuation across 16 pressure zones; validated layout for signal integrity and low-profile form factor
- Implemented and tested real-time pressure heat map in C++ and Python, processing 16-sensor array data to visualize localized pressure distribution and time-rate changes

PROJECTS

RC Car

Jan. 2026 – May 2026

- Designed and prototyped a multi-speed RC car with cam-actuated sliding mesh spur gear transmission — high-speed (1:3), high-torque (9:1), and reverse — controlled via IR remote
- Performed FEA on the TPU impact roof; validated structural integrity with Impact Factor of 63.25 and safety factor of 17.15 under worst-case 3 kg drop from 0.25 m
- Manufactured and assembled drivetrain components via 3D printing and hand tooling: gears, motor mounts, servo mounts, and brake system

Cough-E

Aug. 2025 – Dec. 2025

- Performed PVA analysis on six-bar linkage locomotion system to determine stride length (4.375 cm) and derive the 300:1 compound gear ratio to synchronize a geneva-driven dispenser at 1 payload per 2 m
- Evaluated Delrin vs. $\frac{1}{4}$ " acrylic for linkage fabrication through physical testing; selected acrylic to reduce lateral instability, then laser cut and assembled final linkage components

Canopy Nest

Aug. 2024 – Dec. 2024

- Produced full CAD assembly and GD&T engineering drawings; performed DFA analysis identifying part count reduction from 10 to 2 theoretical minimum through geometry consolidation and process substitution
- Selected materials and manufacturing methods (porcelain slip casting, steel roll forming) through DFM analysis balancing outdoor durability, corrosion resistance, and cost; design reduced projected waste by 30%

TECHNICAL SKILLS

Design & Analysis: SolidWorks, OnShape, Fusion360, KiCAD, FEA, DFM, GD&T, MATLAB/Simulink

Manufacturing & Prototyping: CNC Machining, Injection Molding, Laser Cutting, 3D Printing (FDM), Soldering

Programming & Embedded: C/C++, Python, Arduino, Embedded Systems, Data Acquisition, MATLAB, SQL